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Dynamic electric arc model for electronic circuit simulation

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Analogue-sample: This instruction allows the data on the analogue input pins or test points of a chip to be sampled and placed in the analogue BS cells during normal operation by turning on switches S2 and S4 of all cells.

Table 1: Switch states for input and output analogue boundary scan cells

Input ABSC								
	S1	S2	S3	S4	S5	S6	Enable signal	Comments
Analogue sample	On	On	Off	On	Off	Off	Capture_DR	Sample data
		Off	Off	Off	On	On	dff=1, Shift_DR	Scan out held data
Analogue preload	On	Off	Off	On	Off	On	dff=1	Scan in test data
Analogue intest	Off	Off	On	Off	On	Off	dff=0	DC testing
		Off	On	Off	Off	On	dff=1	AC testing
Analogue extest	Off	On	Off	On	Off	Off	Capture_DR, bsr	DC testing
		On	Off	Off	Off	On	dff=1, $\overline{\text{Shift_dr}}$,bsr	AC testing

Output ABSC								
	S1	S2	S3	S4	S5	S6	Enable signal	Comments
Analogue sample	On	On	Off	On	Off	Off	Capture_DR	Sample data
		Off	Off	Off	On	On	dff=1, Shift_DR	Scan out held data
Analogue preload	On	Off	Off	On	Off	On	dff=1	Scan in test data
Analogue intest	Off	On	Off	On	Off	Off	Capture_DR	DC testing
		On	Off	Off	Off	On	dff=1, $\overline{\text{Shift_dr}}$	AC testing
Analogue extest	Off	Off	On	Off	On	Off	dff=0, bsr	DC testing
		Off	On	Off	Off	On	dff=1	AC testing

Analogue-preload: This instruction is used to prestore test data into analogue BS cells via analogue test buses by turning on the S4 and S6 switches of all analogue cells, one by one.

Analogue-intest: This instruction is used to apply DC test signals to a CUT via input analogue BS cells and capture the responses from the CUT by output analogue BS cells, or to apply AC test signal to CUT via AITB bus and observe responses via AOTB bus. Before executing the analogue-intest instruction, the analogue-preload instruction must be executed to load test data into input BS cells.

Analogue-extest: This instruction is used to test the external nets between two chips. It can also be used to perform DC or AC testing. The test data must be loaded into the analogue boundary scan cells by using the analogue-preload instruction before executing this instruction.

Results and conclusions: According to the above description, some logic is designed to generate the control signals for the ABSCs. Table 1 describes the states of switches for the defined test instructions. The control circuits for the switches of the analogue boundary scan cells have been designed and the simulations of the analogue BS cell and a whole analogue BS architecture have been performed by HSPICE and the Synopsys Synthesis tool, respectively. All operations have been demonstrated properly. With our new architecture the analogue boundary scan methodology can be implemented easily.

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Dynamic electric arc model for electronic circuit simulation

K.J. Tseng and Y. Wang

Indexing terms: Power electronics, Electric breakdown of gases, Discharges (electric)

The authors present a physics based electric arc model which can simulate both static and dynamic characteristics of an arc load controlled by an electronic circuit. The proposed model was developed from modifications of the classical Cassie and Mayr equations. It has been test implemented in the Saber circuit simulator. There has been a considerable degree of correspondence between the simulated and experimental results.

Introduction: To simulate the operation of a power electronic circuit controlling an electric arc load such as an arc discharge lamp or arc plasma reactor, a suitable model for the arc load is required. Owing to the very complex arc phenomena, the relationship between the arc voltage and the arc current, known as arc characteristics, is highly nonlinear. Initially the nonconducting gas presents a very high resistance, followed by a negative resistance during gas breakdown, and followed finally by a small positive resistance once a stable arc is formed. Circuit simulators widely used in the industry and academic research centres, such as SPICE and Saber do not incorporate any arc model. Therefore it is necessary to develop an arc model capable of simulating both static and dynamic behaviour of electric arcs for use in circuit simulators.

A literature survey of arc modelling work carried out so far indicated that most models published were not intended for transient analysis in circuit simulation. These models were postulated to explain certain aspects of the arc phenomena, and they require numerical solution of the arc conservation equations in full differential form [1]. These equations are extremely difficult and costly in terms of computing effort and time, and the required parameters are not easy to obtain. A SPICE circuit model has been proposed recently to simulate a simplified voltage-current characteristic of an arc discharge during the ignition phase. This model is based on a two transistor sub circuit which is therefore divorced from any considerations of arc physics. It is limited to modelling only the ignition phase of an arc and is hence not entirely representative of the dynamic characteristics of an actual arc [2].

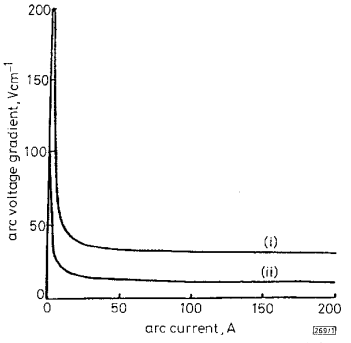


Fig. 1 *Static v-i characteristics of arcs in forced and free convection*
(i) arc in blast, (ii) arc in still air

Arc characteristics: During steady state conditions, any change of arc current takes place slowly. Owing to the fact that increasing current flowing in an arc will tend to increase the ionisation of the

gas, the arc has negative resistance property. Fig. 1 shows typical arc characteristics in a quasi steady state condition. The shape of the curve depends on the mode of heat loss from the arc [3]. Consider a very simple arc model with a constant current density and a constant heat loss per unit area of the cylindrical surface of the arc: the circumference of the arc column, and therefore the heat loss vi (v arc voltage, i arc current) is then proportional to the arc radius. For a constant current density, arc radius is proportional to the square root of arc current. Hence, the relationship between v, i and the arc conductance G is

$$v \propto \frac{1}{\sqrt{i}} \quad \text{and} \quad G \propto i^2 \tag{1}$$

This model gives a shape not unlike the typical characteristics in the low current range, but not the high current range.

In the high current range, we can assume that the current density is constant, and that the heat loss is proportional to the cross-sectional area of the arc, such as might occur with gas flowing through the arc and removing heat from the entire volume of the arc. In this ease, v remains constant, the arc conductance behaves as

$$G \propto i \tag{2}$$

and the model presents an acceptable reproduction of the curve in the high current region.

In practice, the heat loss processes are much more complex. Moreover, the above static arc models are valid only when the changes of current occur slowly. If the arc current changes abruptly, the arc exhibits a hysteresis effect i.e. a time lag in the change in voltage compared to a change in current. This is because the arc plasma conductivity is a function of the gas temperature. Hence, when the current undergoes a positive rate of change, the conductance does not change instantaneously since the temperature change has a time lag owing to the heat capacity of the gas. The result is a lower conductivity and a higher voltage drop than the steady state voltage for a given current.

Proposed arc model: Careful studies, both theoretical and experimental, have shown that the dynamic arc plasma phenomena are extremely complex in detail, so that complete mathematical models of the arc comparable in simplicity and accuracy with engineering models of the connected power electronic circuit are not possible. However, it has been recognised for many years that the relatively simple Cassie and Mayr differential equation models based on simplifications of principal power loss mechanisms and energy storage in the arc column, are useful for at least a qualitative understanding of the phenomena determining arc striking or extinction of the energy balance type [4]. An attempt is made here to modify and combine the Cassie and Mayr models to a form suitable for transient circuit simulation.

Cassie postulated an arc model that has a constant current density, so that its area varied directly with the current [5]. It also has a constant resistivity and stored energy per unit volume. The air (or other materials) flow penetrated the whole cross-section of the arc, carrying away heat, and thus made the dissipation per unit volume constant as well. With the arc resistance R as the dependent variable, the Cassie equation is

$$\frac{1}{R}\frac{dR}{dt} = \frac{1}{\theta}\left(1 - \frac{v^2}{E_0^2}\right) \tag{3}$$

where E_0 is a momentarily constant steady-state arc voltage and θ is the arc time constant = energy stored per unit volume/energy loss rate per unit volume. This model has a drawback in that the modelled arc could not be interrupted. In fact it describes the behaviour of the arc when the current is large.

In Mayr's model, the heat loss is assumed to occur from the periphery of the arc only, and the conductance of the arc varied with the energy stored in it [6]. The Mayr equation is

$$\frac{1}{R}\frac{dR}{dt} = \frac{1}{\theta}\left(1 - \frac{vi}{P_0}\right) \tag{4}$$

where P_0 is a momentarily constant power loss. In the steady-state condition, when currents and voltages are changing very slowly, $vi = P_0$. Thus the steady-state characteristics are hyperbolic, which is a moderately good representation of the curves in the low current range (Fig. 1). This equation does allow the arc to interrupt, since, with R high, vi/P_0 can still be < unity, so that dR/dt is positive and the resistance continues to increase towards the final interruption.

In transient circuit simulations using an iterative Newton-Raphson algorithm, convergence problems are encountered whenever the value of R becomes very small during any of the iterations. Therefore we propose using the arc conductance G as the dependent variable instead of R. It can be shown that the Cassie equation can be transformed to

$$G = (vi/E_0^2) - \theta(dG/dt) \tag{5}$$

and the Mayr equation transformed to

$$G = (i^2/P_0) - \theta(dG/dt) \tag{6}$$

Eqns. 5 and 6 are consistent with the corresponding static relationships given by eqns. 2 and 1, respectively. As the Mayr equation is more valid for zero and low current regions and the Cassie equation is more representative of higher current regions, it is proposed here that an arc may be simulated by the combination of eqns. 5 and 6. It is assumed here that the value of arc time constant θ is common to both equations.

To allow smooth transition between eqns. 5 and 6 during a transient simulation, the following transition factors are proposed for the static terms of the equations:

$$\text{for transformed Cassie equation: } \left[1 - \exp(-i^2/I_0^2)\right] \tag{7}$$

$$\text{for transformed Mayr equation: } \left[\exp(-i^2/I_0^2)\right] \tag{8}$$

where I_0 is the transition current. The advantage of this form of transition is the elimination of the lack of a defined derivative at the transition point between the two equations.

In addition, there has to be a finite, though very small, amount of conductance between two electrodes when the arc is absent. The value of this constant G_{min} would depend on the distance between the electrodes, the geometry of the electrodes, the type of gas, and the temperature. The complete model is thus given by

$$G = G_{min} + \left[1 - \exp\left(\frac{i^2}{I_0^2}\right)\right]\frac{vi}{E_0^2} + \left[\exp\left(\frac{i^2}{I_0^2}\right)\right]\frac{i^2}{P_0} - \theta\frac{dG}{dt} \tag{9}$$

and

$$i = Gv \tag{10}$$

Experimental verification: To test the validity of the model, we have implemented it in the Saber simulator, one of the widely used circuit simulators in the industry. The models that describe each of the components of the simulated system can be accessed from the Saber libraries of standard component models, or from user defined Saber templates where the equations that describe the physical behaviour of the device are implemented. Saber templates arc written in the MAST modelling language, which is similar to the C programming language with the addition of specially designed modelling constructs which facilitate the implementation of Kirchhoff's laws and aid convergence [7]. The arc model has been implemented in the MAST language as a two terminal device whose current and voltage are given by ens. 9 and 10.

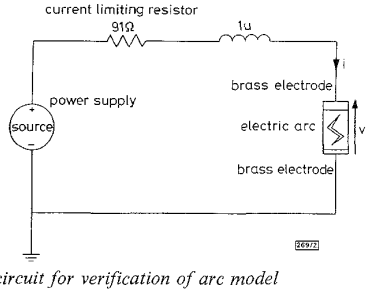


Fig. 2 *Test circuit for verification of arc model*

The arc model was evaluated using the test circuit shown in Fig. 2. Two brass electrodes 3mm in diameter were used to create the arc. The arc was ignited by bringing the electrodes together and then slowly separating them to form an arc length of 5mm. To measure its static characteristics a DC power supply was used. The DC supply voltage was gradually reduced while recording the arc voltage and current until the arc was extinguished. For the investigation of dynamic characteristics, a 50Hz AC power supply of $415\,V_{rms}$ was used so that the instantaneous relationship between the arc voltage and current can be observed and captured on a digital oscilloscope.

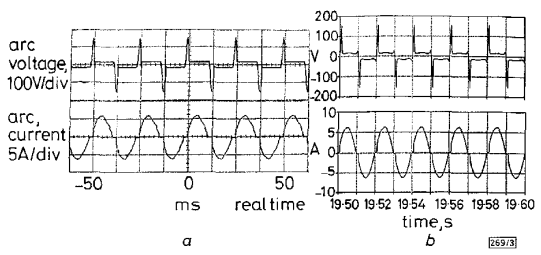


Fig. 3 *Comparison of simulated and experimental waveforms of arc voltage and current*

a Experimental oscillogram
b Simulated using Saber

By trial and error, the model parameters for this particular arc setup have been determined as: $E_0 = 28$, $I_0 = 4.8$, $P_0 = 30$, $G_{min} = 1 \times 10^{-8}$ and $\theta = 1 \times 10^{-4}$. Fig. 3 shows the simulated and experimental results for the instantaneous voltage and current waveforms when the arc is excited by an AC supply. It can be seen that for each half-cycle (10ms) of the AC supply voltage, the arc is momentarily extinguished and then re-ignited. The figures show good correlation between the simulated and experimental results.

Conclusion: An electric arc model is proposed which facilitates the transient simulation of electronic circuits controlling electric arc loads. The model has its roots in the classical Cassie and Mayr equations, which were used extensively in the past to explain the physics of arc plasma operation. The equations of the proposed model have been cast in a form suitable for implementation in industrial-standard circuit simulators. Experimental results obtained so indicated that the model is able to predict the static and dynamic characteristics of electric arcs with reasonable accuracy.

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Efficient algorithm for untestable path detection

Hun Heo and Sun Young Hwang

Indexing terms: Digital circuits, Logic circuits, Circuit testing

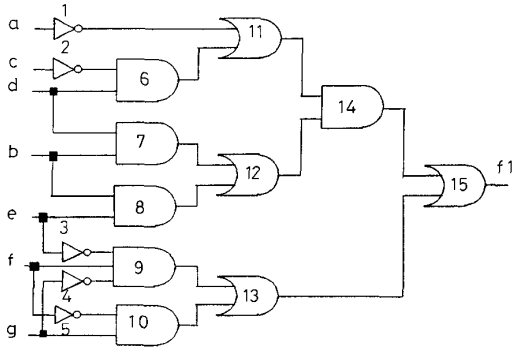
The authors present an algorithm for detecting untestable paths in multi-level circuits. By constructing equivalent normal form (ENF) for reconvergent paths only, the proposed algorithm detects and removes untestable paths efficiently in terms of runtime and memory usage.

Introduction: Recently, the need for ensuring the temporal correctness of a circuit has led to the development of delay fault testing. The path delay fault model is widely adopted for delay fault testing since it can give guarantees on circuit performance [1, 2]. However, path-delay fault testing often requires a large number of test vectors and a long CPU time in test generation, since the number of paths is extremely large even in moderate-sized circuits.

Testable logic synthesis has been proposed as a possible solution for overcoming this problem [1]. The testable logic synthesis improves the testability of a circuit by removing untestable paths which do not allow test sets for detecting path delay faults.

Previous work in path delay faults has detected untestable paths in multi-level circuits using the analysis [3]. The ENF of a circuit is a two-level representation, in which cubes represent sensitisation conditions for each path. Each literal in the cube has a tag representing its path from primary inputs to a primary output. However, in constructing the ENF of a multi-level circuit, huge amounts of CPU time and memory are used in the worst case [3].

In this Letter an efficient algorithm for detecting untestable paths is described. The proposed algorithm efficiently detects untestable paths by exploiting the characteristics of untestable paths. They are reconvergent paths and have the same parity.



a

Cube 1 : $a'\{1,11,14,15\}*b\{7,12,14,15\}*d\{7,12,14,15\}$
Cube 2 : $a'\{1,11,14,15\}*b\{8,12,14,15\}*e\{8,12,14,15\}$
Cube 3 : $b\{7,12,14,15\}*c'\{2,6,11,14,15\}*d\{6,11,14,15\}*d\{7,12,14,15\}$
Cube 4 : $b\{8,12,14,15\}*c'\{2,6,11,14,15\}*d\{6,11,14,15\}*e\{7,12,14,15\}$
Cube 5 : $e'\{3,9,13,15\}*f\{9,13,15\}*g'\{4,9,13,15\}$
Cube 6 : $f'\{5,10,15\}*g\{10,15\}$

b

Cube 7 : $a'\{1,11,14\} \cdot b\{7,12,14\} \cdot d\{7,12,14\}$
Cube 8 : $a'\{1,11,14\} \cdot b\{8,12,14\} \cdot e\{9,12,14\}$
Cube 9 : $b\{7,12,14\} \cdot c'\{2,6,11,14\} \cdot d\{8,11,14\} \cdot d\{7,12,14\}$
Cube 10 : $b\{8,12,14\} \cdot c'\{2,8,11,14\} \cdot d\{8,11,14\} \cdot e\{7,12,14\}$

c

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Fig. 1 *ENF expressions for primary outputs and for reconvergent paths*

a An example circuit
b ENF expressions for primary outputs
c ENF expressions for reconvergent paths

ENF analysis in previous works: Previous works detect and remove untestable paths in multi-level circuits by constructing ENF for each primary output of a circuit. After the logic cone for each output is identified and each complex gate in the logic cone is decomposed into AND gates, OR gates and inverters, the ENF is constructed for each primary output. The ENF expression for each primary output represents a set of cubes without performing Boolean identity operations. Each literal in the cube has a tag regarding its path from primary inputs to a primary output.

In Fig. 1 we show the ENF expressions for the primary output f1 and for the reconvergent gate 14. The ENF expression for the AND gate 6 is $c'\{2,6\}*d\{6\}$. The ENF for the AND gate 14 is $a'\{1,11,14\}*b\{7,12,14\}*d\{7,12,14\} + a'\{1,11,14\}*b\{8,12,14\}*e\{8,12,14\} + c'\{2,6,11,14\}*b\{7,12,14\}*d\{5,11,14\}*d\{7,12,14\} + c'\{2,6,11,14\}*b\{8,12,14\}*d\{5,11,14\}*e\{9,12,14\}$.

The constructed ENFs contain information on what test vector can set a gate output to 0/1. Since untestable paths are the paths that no test vectors can sensitise alone, they can be easily identified through the constructed ENF. If a test vector sensitises only a single path, they can detect the path delay faults along the path.